Core 1 Day 9 Hardware

Network Cables

Transmission Speeds
- ↳ Copper cables → 40 Gb upto
- ↳ Fiber cables → 100 Gb

Transmission Distance
- ↳ Copper Cables → 1100 meters (3609 feet)
- ↳ Fiber Cables → 40 km (25 miles)

Analog telephone connector
↳ RJ 11

Twisted Pair Connector

RJ 11 : 4 pin connector, smaller

RJ45 : 8 pin connector
↳ found in computer networks, wider

Twisted Pair Categories

Category	Speed	Distance	Note
cat5	100Mbps	100 meters	older networks
cat5e	1000 Mbps/ 1Gbps	100meters	More twisted per foot
Cat 6	10000Mbps/ 10Gbps 1000 Mbps 1 Gbps	55meters 100 meters	Includes piece of plastic to minimize crosstalk.
Cat 6a	10000 Mbps/ 10Gbps	100 meters	Thicker wires

Cat7

- Attenuation
  is the loss of signal strength
- Noise Immunity
  - ↳ EMI (Electro-Magnetic Interference)
    - ↳ leak or disrupt signals
  - ↳ Copper cables
    - ↳ Highly susceptible to interference.
      use shielded cables to protect against EMI
  - ↳ Fiber Cables
    are NOT susceptible to EMI since it is not Copper.

Wiring Standards (two)
↳ RJ45 pinouts
(T568 A and T568 B) standards
- ↳ The blue and brown pairs are always in the same location
- ↳ only orange and green pairs transpose their position

Twisted Pair Cables ☆☆☆
↳ 8 cables inside
↓
4 pairs
most used cable
two types
- ↳ STP (Sheild Twisted Pair)
  - ↳ agaist EMI
  - ↳ • Direct-burial cable
- ↳ UTP (Unshield Twisted Pair)
  - ↳ no protection against EMI

Coaxial Cable
↳ cable in or satellite television connections
- RG6 most common type

Advantage → Sheilded → EMI
→ Long transmission distance (1100 meters)
→ affordable than fiber optic

Disadvantage → More expensive than twisted pair
→ Copper Core can be snap if mishandled

Coax Connectors
- BNC (old)
- F connector